

Free-Field Computation of the $\mathbf{Vir} \oplus \mathbf{Vir}$ Branching Coefficient

Highest-vector prescription, norm, and three-point numerator

Machine working note

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Abstract

The human note reduces the all-NS superconformal block to two ordinary Virasoro blocks and leaves one quantity undetermined: the normalized three-point coefficient of the $\mathbf{Vir} \oplus \mathbf{Vir}$ highest-weight vectors v_n . We use the free-field highest-vector prescription suggested by arXiv:1111.2803, but every norm and three-point coefficient is evaluated with the BPZ, tensor-product, and fixed-parity conventions of the human note. Writing $k = 2n \in \mathbb{Z}$, the vector is a finite Fermi-sea product of the modes $\chi_r = \psi_r - i\eta_r$. The factorized ratio l in the paper is recorded only as a comparison; it is not identified with the human-note three-point function. The complete direct calculation at $n_i \in \{0, \pm\frac{1}{2}\}$ is given below. No Whittaker vector is used.

Contents

1	NS free-field realization	2
1.1	Oscillators and Fock vacuum	2
1.2	SCA generators	3
1.3	The commuting Virasoro pair	3
2	The quantity left open in the human note	4
3	Highest vector from the free-field realization	4
3.1	The Fermi-sea vector	4
4	Factorized three-point numerator	5
4.1	Dictionary for an ordered trinion	5
4.2	The product formula	6
4.3	Endpoint order and the first convention check	7
5	Norm from the same formula	7
6	Squared coefficient obtained from the literature ratio	8

7	Low-level values	8
7.1	Complete unpacking for $n_i = 0, \pm \frac{1}{2}$	9
8	Constructive physical-L_1 action: NS sector	11
8.1	The embedded generators, with the ψG term displayed	11
8.2	Complex-fermion charge and the pair-field identity	11
8.3	Explicit Heisenberg descendant and normalization	13
8.4	Triangular J_0 filtration and support theorem	13
8.5	Relative current–PBW reconstruction in the human-note conventions	15
8.6	First exact NS checks	17
9	Ramond extension: status of the literature	18
9.1	What has actually been established	18
10	Constructive physical-L_1 action: Ramond sector	19
10.1	Selection rule and exact low-level reductions	20
10.2	Constructive free-field proof of the current-descendant formula	21
10.3	Triangular J_0 filtration and support theorem	25
11	Ramond literature verdict and remaining problem	26
11.1	Verdict	26
11.2	Screening-charge route to be pursued here	27
12	What remains to be audited	27
13	Sources	28

1 NS free-field realization

We first fix the realization in which the branching vector will be computed. Set

$$Q = b + b^{-1}, \quad c = \frac{3}{2} + 3Q^2, \quad h(P) = \frac{1}{2} \left(\frac{Q^2}{4} - P^2 \right). \quad (1)$$

All modes in this section are NS modes: $m \in \mathbb{Z}$ and $r \in \mathbb{Z} + \frac{1}{2}$.

1.1 Oscillators and Fock vacuum

Following the human note, the auxiliary Majorana fermion has modes ψ_r . The free-field realization of the SCA uses a Heisenberg oscillator c_m^ϵ , a physical NS fermion η_r^ϵ , and a bosonic zero mode $\mathcal{P}$.

Here $\epsilon \in \{+1, -1\}$ labels two *alternative reflected realizations* of the same SCA module; the $+$ and $-$ oscillators are not two simultaneous free fields. Their nonzero brackets are

$$\begin{aligned} [c_m^\epsilon, c_n^\epsilon] &= m\delta_{m+n,0}, & \{\eta_r^\epsilon, \eta_s^\epsilon\} &= \delta_{r+s,0}, & \{\psi_r, \psi_s\} &= \delta_{r+s,0}, \\ [\mathcal{P}, c_m^\epsilon] &= [\mathcal{P}, \eta_r^\epsilon] = 0, & [\psi_r, c_m^\epsilon] &= 0, & \{\psi_r, \eta_s^\epsilon\} &= 0. \end{aligned} \quad (2)$$

The auxiliary BPZ convention inherited from the human note is

$$\psi_r^\dagger = -\psi_{-r}. \quad (3)$$

Let $\mathbf{1}_F$ be annihilated by ψ_r for $r > 0$, and let $|P\rangle_\epsilon$ be annihilated by c_m^ϵ and η_r^ϵ for $m, r > 0$, with $\mathcal{P}|P\rangle_\epsilon = P|P\rangle_\epsilon$. We write

$$|\text{vac}; P\rangle := \mathbf{1}_F \otimes |P\rangle_\epsilon. \quad (4)$$

Thus the tensor-product module is the span of arbitrary finite products of the negative modes ψ_{-r} , c_{-m}^ϵ , and η_{-r}^ϵ acting on $|\text{vac}; P\rangle$.

1.2 SCA generators

In the realization labelled by ϵ , the abstract SCA generators act for $m \neq 0$ as

$$\begin{aligned} L_m &= \frac{1}{2} \sum_{j \neq 0, m} c_j^\epsilon c_{m-j}^\epsilon + \frac{1}{2} \sum_{r \in \mathbb{Z} + \frac{1}{2}} \left(r - \frac{m}{2}\right) \eta_{m-r}^\epsilon \eta_r^\epsilon + \frac{i}{2} (Qm + 2\epsilon\mathcal{P}) c_m^\epsilon, \\ G_r &= \sum_{m \neq 0} c_m^\epsilon \eta_{r-m}^\epsilon + i(Qr + \epsilon\mathcal{P}) \eta_r^\epsilon, \\ L_0 &= \sum_{m > 0} c_{-m}^\epsilon c_m^\epsilon + \sum_{r > 0} r \eta_{-r}^\epsilon \eta_r^\epsilon + \frac{1}{2} \left(\frac{Q^2}{4} - \mathcal{P}^2 \right). \end{aligned} \quad (5)$$

These formulas obey the $\mathcal{N} = 1$ SCA relations at the central charge in Eq. (1), and $L_0|P\rangle_\epsilon = h(P)|P\rangle_\epsilon$. The realization $\epsilon = -1$ is used for $k > 0$, and $\epsilon = +1$ for $k < 0$. They are related by the super-Liouville reflection operator.

It is now meaningful to define the isotropic fermionic modes

$$\chi_r^\epsilon := \psi_r - i\eta_r^\epsilon, \quad \{\chi_r^\epsilon, \chi_s^\epsilon\} = 0. \quad (6)$$

The vanishing anticommutator is the reason that the highest vector below is a finite Fermi-sea product.

1.3 The commuting Virasoro pair

For completeness, define the auxiliary-fermion stress tensor and the mixed bilinear

$$L_n^F = \frac{1}{2} \sum_{r \in \mathbb{Z} + \frac{1}{2}} r : \psi_{n-r} \psi_r :, \quad U_n = \sum_{r \in \mathbb{Z} + \frac{1}{2}} \psi_{n-r} G_r. \quad (7)$$

The two commuting Virasoro generators used in the human note are

$$L_n^{(1)} = \frac{b^{-1}}{b^{-1} - b} L_n - \frac{b^{-1} + 2b}{b^{-1} - b} L_n^F + \frac{1}{b^{-1} - b} U_n,$$

$$L_n^{(2)} = \frac{b}{b-b^{-1}}L_n - \frac{b+2b^{-1}}{b-b^{-1}}L_n^F + \frac{1}{b-b^{-1}}U_n. \quad (8)$$

They obey

$$[L_m^{(1)}, L_n^{(2)}] = 0, \quad c^{(i)} = 1 + 6(Q^{(i)})^2, \quad Q^{(i)} = b^{(i)} + (b^{(i)})^{-1}, \quad (9)$$

where

$$b^{(1)} = \frac{2b}{\sqrt{2-2b^2}}, \quad (b^{(2)})^{-1} = \frac{2b^{-1}}{\sqrt{2-2b^{-2}}}. \quad (10)$$

This finishes the algebraic setup. Every branching vector computed below is an explicit vector in the Fock space of Eqs. (2)–(4).

2 The quantity left open in the human note

The human note uses

$$\mathcal{M}_{\text{NS}}(h) = \mathcal{V}_{F_{1/2}} \otimes \mathcal{V}_{\text{SCA}_{1/2}}(h), \quad h = \frac{1}{2} \left(\frac{Q^2}{4} - P^2 \right), \quad Q = b + b^{-1}, \quad (11)$$

with the auxiliary free-fermion factor first. The branching rule is

$$\mathcal{M}_{\text{NS}}(h) \cong \bigoplus_{n \in \frac{1}{2}\mathbb{Z}} \mathcal{V}_{c^{(1)}}(h_n^{(1)}) \otimes \mathcal{V}_{c^{(2)}}(h_n^{(2)}), \quad h_n^{(1)} + h_n^{(2)} = h + 2n^2. \quad (12)$$

The corresponding highest-weight vector is denoted by v_n , and its parity is $2n \pmod{2}$.

For the ordered trinion used throughout the human note, slot 1 is the BPZ-conjugate state at infinity, slot 2 is inserted at 1, and slot 3 is the ket at 0. In precisely this order the human note defines

$$B_a(h_1, h_2, h_3; n_1, n_2, n_3) = \frac{\hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})}{\|v_{1,n_1}\| \|v_{2,n_2}\| \|v_{3,n_3}\|}, \quad a = 2(n_1 + n_2 + n_3) \pmod{2}. \quad (13)$$

Only B_a^2 enters the conformal-block decomposition. Directly from the human definition,

$$[B_a(h_1, h_2, h_3; n_1, n_2, n_3)]^2 = \frac{\hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})^2}{N_{k_1}(P_1)N_{k_2}(P_2)N_{k_3}(P_3)}, \quad k_i = 2n_i, \quad N_{k_i}(P_i) = \|v_{i,n_i}\|^2. \quad (14)$$

No reversal of the labels $1 \leftrightarrow 3$ is made in this note. Other slot orders must be obtained with the Möbius and Koszul conventions already fixed in the human note.

3 Highest vector from the free-field realization

3.1 The Fermi-sea vector

Set $k = 2n \in \mathbb{Z}$. For $k \neq 0$, the prescription is

$$|P, k\rangle = \Omega_k(P) \prod_{j=1}^{|k|} \chi_{-(2j-1)/2}^{-\text{sgn}(k)} |\text{vac}; P\rangle, \quad |P, 0\rangle = |\text{vac}; P\rangle. \quad (15)$$

The product is ordered with increasing j from left to right. Its total level is

$$\sum_{j=1}^{|k|} \frac{2j-1}{2} = \frac{k^2}{2} = 2n^2, \quad (16)$$

which is precisely the shift in Eq. (12).

The factor $\Omega_k(P)$ is fixed by the standard SCA-PBW normalization

$$|P, k\rangle = \left(G_{-1/2}^{k^2} + \text{terms of lower degree in } G \right) |P\rangle. \quad (17)$$

This fixes the vector, including its scale, without referring to a Whittaker state.

The commutators quoted in the literature are

$$\begin{aligned} [L_m^{(1)} + L_m^{(2)}, \chi_r^\epsilon] &= -\left(\frac{m}{2} + r\right) \chi_{m+r}^\epsilon, \\ [bL_m^{(1)} + b^{-1}L_m^{(2)}, \chi_r^\epsilon] &= -((m+r)Q + \epsilon\mathcal{P}) \chi_{m+r}^\epsilon + i \sum_{j \neq 0} c_j^\epsilon \chi_{m+r-j}^\epsilon. \end{aligned} \quad (18)$$

They prove directly that Eq. (15) is killed by both positive Virasoro algebras. The two weights are

$$\begin{aligned} h_n^{(1)} &= \frac{1}{4}(Q^{(1)})^2 - \left(\frac{P}{\sqrt{2-2b^2}} + nb^{(1)}\right)^2, \\ h_n^{(2)} &= \frac{1}{4}(Q^{(2)})^2 - \left(\frac{P}{\sqrt{2-2b^{-2}}} + n(b^{(2)})^{-1}\right)^2. \end{aligned} \quad (19)$$

The first nontrivial vectors, already converted back to the human note's SCA and auxiliary-fermion modes, are

$$\begin{aligned} |P, 1\rangle &= \left[G_{-1/2} + \left(\frac{Q}{2} + P\right) \psi_{-1/2} \right] |P, 0\rangle, \\ |P, -1\rangle &= \left[G_{-1/2} + \left(\frac{Q}{2} - P\right) \psi_{-1/2} \right] |P, 0\rangle. \end{aligned} \quad (20)$$

They are the human-note vectors $v_{1/2}$ and $v_{-1/2}$.

4 Factorized three-point numerator

4.1 Dictionary for an ordered trinion

Parameterize the inserted NS weight by

$$h_2 = \frac{1}{2}\alpha_2(Q - \alpha_2), \quad \alpha_2 = \frac{Q}{2} + P_2. \quad (21)$$

For the ordered matrix element

$$\langle P_1, k_1 | \mathbb{V}_{\alpha_2}^{(k_2)}(1) | P_3, k_3 \rangle, \quad (22)$$

the state-field correspondence identifies $\mathbb{V}_{\alpha_2}^{(k_2)}$ with the same highest vector $|P_2, k_2\rangle$ used on the middle leg. The normalized ratio in arXiv:1111.2803 is

$$l(\alpha, m \mid P', k', P, k) = \begin{cases} \frac{\langle P', k' | \mathbb{V}_{\alpha}^{(m)} | P, k \rangle}{\langle P' | \mathbb{V}_{\alpha}^{(0)} | P \rangle}, & m + k + k' \text{ even}, \\ \frac{\langle P', k' | \mathbb{V}_{\alpha}^{(m)} | P, k \rangle}{\langle P' | \mathbb{V}_{\alpha}^{(\pm 1)} | P \rangle}, & m + k + k' \text{ odd}. \end{cases} \quad (23)$$

The corresponding primary denominators have the same numerical normalization as the human note: $\rho_0(\phi_1, \phi_2, \phi_3) = 1$ in the even case and $\rho_1(\phi_1, G_{-1/2}\phi_2, \phi_3) = 1$ in the odd case. Define the literature ratio in the human slot labels by

$$L_a^{\text{lit}}(1, 2, 3) := l(\alpha_2, k_2 \mid P_1, k_1, P_3, k_3), \quad a = k_1 + k_2 + k_3 \pmod{2}. \quad (24)$$

Matching the two primary denominators does *not* by itself prove

$$T_a^{\text{H}} = L_a^{\text{lit}}(1, 2, 3). \quad (25)$$

Equation (25) is a convention-sensitive statement about the complete highest vectors. It must also match the auxiliary fermion BPZ sign, the fixed-parity conversion in the human note, and the state-field map for the middle vector. This is the central audit below.

4.2 The product formula

For $\sigma, \tau \in \{+1, -1\}$, define

$$x_{\sigma\tau} := \alpha + \sigma P' + \tau P, \quad r_{\sigma\tau} := \frac{m + \sigma k' + \tau k}{2}. \quad (26)$$

For a nonnegative integer r , set

$$\begin{aligned} s_{\text{even}}(x, r) &= 2^{-r^2/2} \prod_{\substack{i, j \geq 1, \ i+j \leq 2r \\ i+j \equiv 0 \pmod{2}}} [x + (i-1)b + (j-1)b^{-1}], \\ s_{\text{odd}}(x, r) &= 2^{-r(r+1)/2} \prod_{\substack{i, j \geq 1, \ i+j \leq 2r+1 \\ i+j \equiv 1 \pmod{2}}} [x + (i-1)b + (j-1)b^{-1}]. \end{aligned} \quad (27)$$

For negative integer r , extend them by

$$s_{\text{even}}(x, r) = (-1)^r s_{\text{even}}(Q - x, -r), \quad s_{\text{odd}}(x, r) = s_{\text{odd}}(Q - x, -r). \quad (28)$$

Finally define

$$\text{Int}(x) := \text{sgn}(x) \lfloor |x| \rfloor. \quad (29)$$

The literature ratio is then

$$l(\alpha, m \mid P', k', P, k) = \begin{cases} \prod_{\sigma, \tau = \pm 1} s_{\text{even}}(x_{\sigma\tau}, r_{\sigma\tau}), & m + k + k' \text{ even}, \\ \prod_{\sigma, \tau = \pm 1} s_{\text{odd}}(x_{\sigma\tau}, \text{Int}(r_{\sigma\tau})), & m + k + k' \text{ odd}. \end{cases} \quad (30)$$

This is the factorized matrix-element ratio of arXiv:1111.2803. It becomes the desired human-note numerator only after Eq. (25), including any convention factor, has been established.

4.3 Endpoint order and the first convention check

The fixed-parity trilinear form used in the human note obeys the endpoint reflection rule

$$\widehat{\rho}_a(x_1, x_2, x_3) = (-1)^{|x_2|(|x_1|+|x_3|)} \widehat{\rho}_a(x_3, x_2, x_1), \quad (31)$$

with the weights and labels in slots (1) and (3) exchanged on the right-hand side. For the branching primaries this sign is $(-1)^{k_2(k_1+k_3)}$. Thus the reversed expression $\widehat{\rho}_a(v_{3,n_3}, v_{2,n_2}, v_{1,n_1})$, used in an earlier version of this note, is not the definition in the human note and cannot be substituted without this reflection sign.

Already at $n_i = 0, \pm \frac{1}{2}$, one must expand the explicit vectors

$$v_{i,\epsilon_i/2} = G_{-1/2}\phi_i + a_i\psi_{-1/2}\phi_i, \quad a_i := \frac{Q}{2} + \epsilon_i P_i, \quad \epsilon_i = \pm 1, \quad (32)$$

inside the tensor-product trilinear form of the human note. Its auxiliary BPZ convention gives

$$\rho^F(\psi\mathbf{1}, \psi\mathbf{1}, \mathbf{1}) = -1, \quad \rho^F(\psi\mathbf{1}, \mathbf{1}, \psi\mathbf{1}) = -1, \quad \rho^F(\mathbf{1}, \psi\mathbf{1}, \psi\mathbf{1}) = +1. \quad (33)$$

Using the human note's fixed-parity component table then gives

$$\begin{aligned} T_0^H[\epsilon_1/2, \epsilon_2/2, 0] &= -\frac{1}{2} \left[\left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 \right)^2 - P_3^2 \right], \\ T_0^H[\epsilon_1/2, 0, \epsilon_3/2] &= \frac{1}{2} \left[P_2^2 - \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_3 P_3 \right)^2 \right], \\ T_0^H[0, \epsilon_2/2, \epsilon_3/2] &= \frac{1}{2} \left[\left(\frac{Q}{2} + \epsilon_2 P_2 + \epsilon_3 P_3 \right)^2 - P_1^2 \right]. \end{aligned} \quad (34)$$

The product formula for L^{lit} has the same last two entries but the opposite sign in the first entry. Equivalently, on these cases,

$$T_a^H = (-1)^{k_1 k_2} L_a^{\text{lit}}. \quad (35)$$

This establishes a genuine convention factor at the first nontrivial level. Whether Eq. (35) is the complete all-level translation must be checked with an explicit $|k| = 2$ vector; it is not assumed below for an unsquared coefficient.

5 Norm from the same formula

Take the inserted field to be the identity:

$$\alpha = 0, \quad m = 0, \quad P' = P, \quad k' = k. \quad (36)$$

Then Eq. (23) becomes the norm, while the four factors in Eq. (30) reduce to

$$(x_{++}, r_{++}) = (2P, k), \quad (x_{+-}, r_{+-}) = (0, 0), \quad (x_{-+}, r_{-+}) = (0, 0), \quad (x_{--}, r_{--}) = (-2P, -k). \quad (37)$$

Since $s_{\text{even}}(x, 0) = 1$, we obtain

$$\boxed{N_k(P) = \langle P, k | P, k \rangle = s_{\text{even}}(2P, k) s_{\text{even}}(-2P, -k).} \quad (38)$$

Equivalently, for $p \geq 0$,

$$\begin{aligned} N_p(P) &= (-1)^p s_{\text{even}}(2P, p) s_{\text{even}}(Q + 2P, p), \\ N_{-p}(P) &= (-1)^p s_{\text{even}}(Q - 2P, p) s_{\text{even}}(-2P, p). \end{aligned} \quad (39)$$

At level $1/2$, Eq. (20) gives an independent direct check. Using

$$\langle G_{-1/2}\phi, G_{-1/2}\phi \rangle = 2h, \quad \langle \psi_{-1/2}\mathbf{1}, \psi_{-1/2}\mathbf{1} \rangle = -1, \quad (40)$$

and vanishing mixed inner product, one finds

$$\begin{aligned} N_1(P) &= 2h - \left(\frac{Q}{2} + P\right)^2 = -P(Q + 2P), \\ N_{-1}(P) &= 2h - \left(\frac{Q}{2} - P\right)^2 = P(Q - 2P). \end{aligned} \quad (41)$$

These agree with Eq. (38), because $s_{\text{even}}(x, 1) = x/\sqrt{2}$. The same computation also gives

$$\langle P, 1 | P, -1 \rangle = 0. \quad (42)$$

6 Squared coefficient obtained from the literature ratio

The product formula and the norm determine the following literature-normalized candidate:

$$\mathcal{B}_a^{\text{lit}}(P_1, P_2, P_3; k_1, k_2, k_3) = \frac{l\left(\frac{Q}{2} + P_2, k_2 \mid P_1, k_1, P_3, k_3\right)^2}{N_{k_1}(P_1)N_{k_2}(P_2)N_{k_3}(P_3)}, \quad (43)$$

where l is Eq. (30), N is Eq. (38), and

$$k_i = 2n_i, \quad a = k_1 + k_2 + k_3 \pmod{2}. \quad (44)$$

At the cases checked in Eq. (34), the translation is a sign and hence $\mathcal{B}_a^{\text{lit}} = \mathcal{B}_a$. The all-level identification requires proving that the translation between T_a^{H} and L_a^{lit} remains a pure sign for the explicit higher branching vectors.

If an unsquared B_a is desired, one must choose square roots of the three complex Shapovalov norms, fix an overall phase for every branch vector, and include the human-to-literature trilinear convention factor. Principal square roots alone do not define the human-note coefficient.

7 Low-level values

The product formula immediately gives

$$l(\alpha, 0 \mid P', 0, P, 0) = 1, \quad l(\alpha, 1 \mid P', 0, P, 0) = 1. \quad (45)$$

The first is the even primary normalization; the second is the odd normalization.

With one $k = 1$ branch vector and the other two labels zero,

$$l(\alpha, 0 \mid P', 0, P, 1) = l(\alpha, 0 \mid P', 1, P, 0) = 1. \quad (46)$$

Consequently the corresponding squared normalized coefficient is simply the inverse norm of that leg.

With two $k = 1$ vectors, the even numerators are

$$\begin{aligned} l(\alpha, 0 \mid P', 1, P, 1) &= \frac{1}{2}(\alpha + P' + P)(\alpha - P' - P - Q), \\ l(\alpha, 1 \mid P', 1, P, 0) &= \frac{1}{2}(\alpha + P' + P)(\alpha + P' - P), \\ l(\alpha, 1 \mid P', 0, P, 1) &= \frac{1}{2}(\alpha + P' + P)(\alpha - P' + P). \end{aligned} \quad (47)$$

These are the first nontrivial polynomial checks to reproduce directly from the human-note Ward identities.

For example,

$$\mathcal{B}_0^{\text{lit}}(P_1, P_2, P_3; 1, 0, 1) = \frac{(\alpha_2 + P_3 + P_1)^2(\alpha_2 - P_3 - P_1 - Q)^2}{4N_1(P_3)N_1(P_1)}, \quad \alpha_2 = \frac{Q}{2} + P_2. \quad (48)$$

7.1 Complete unpacking for $n_i = 0, \pm \frac{1}{2}$

We now remove the product notation completely at the first nontrivial branching level. In this subsection the slot order (n_1, n_2, n_3) is exactly the human order: BPZ-conjugate state at infinity, inserted field at one, and ket at zero. To keep the left-hand sides readable, define

$$\mathcal{B}_a^{\text{lit}}[n_1, n_2, n_3] := \mathcal{B}_a^{\text{lit}}(P_1, P_2, P_3; 2n_1, 2n_2, 2n_3), \quad L_a^{\text{lit}}[n_1, n_2, n_3] := l\left(\frac{Q}{2} + P_2, 2n_2 \mid P_1, 2n_1, P_3, 2n_3\right). \quad (49)$$

The human numerator remains $T_a^{\text{H}}[n_1, n_2, n_3] = \hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})$.

Whenever $n_i \neq 0$, write

$$\epsilon_i := 2n_i \in \{+1, -1\}. \quad (50)$$

The free-field vector and its norm take the uniform form

$$\begin{aligned} |P, \epsilon\rangle &= \left[G_{-1/2} + \left(\frac{Q}{2} + \epsilon P \right) \psi_{-1/2} \right] |P, 0\rangle, \\ \langle P, \epsilon | P, \epsilon \rangle &= -\epsilon P (Q + 2\epsilon P). \end{aligned} \quad (51)$$

Zero or one nonzero label. For the vacuum branch,

$$T_0^{\text{H}}[0, 0, 0] = L_0^{\text{lit}}[0, 0, 0] = 1, \quad \mathcal{B}_0^{\text{lit}}[0, 0, 0] = 1. \quad (52)$$

If exactly one label is nonzero, the numerator remains one and

$$\begin{aligned} T_1^{\text{H}}[\epsilon_1/2, 0, 0] &= T_1^{\text{H}}[0, \epsilon_2/2, 0] = T_1^{\text{H}}[0, 0, \epsilon_3/2] \\ &= L_1^{\text{lit}}[\epsilon_1/2, 0, 0] = L_1^{\text{lit}}[0, \epsilon_2/2, 0] = L_1^{\text{lit}}[0, 0, \epsilon_3/2] = 1, \\ \mathcal{B}_1^{\text{lit}}[\epsilon_1/2, 0, 0] &= -\frac{\epsilon_1}{P_1(Q + 2\epsilon_1 P_1)}, \\ \mathcal{B}_1^{\text{lit}}[0, \epsilon_2/2, 0] &= -\frac{\epsilon_2}{P_2(Q + 2\epsilon_2 P_2)}, \\ \mathcal{B}_1^{\text{lit}}[0, 0, \epsilon_3/2] &= -\frac{\epsilon_3}{P_3(Q + 2\epsilon_3 P_3)}. \end{aligned} \quad (53)$$

For example, a $+\frac{1}{2}$ label gives $-[P_i(Q + 2P_i)]^{-1}$, whereas a $-\frac{1}{2}$ label gives $[P_i(Q - 2P_i)]^{-1}$.

Exactly two nonzero labels. When the bra label vanishes, the literature ratio is

$$\begin{aligned} L_0^{\text{lit}}[0, \epsilon_2/2, \epsilon_3/2] &= \frac{1}{2} \left[\left(\frac{Q}{2} + \epsilon_2 P_2 + \epsilon_3 P_3 \right)^2 - P_1^2 \right], \\ \mathcal{B}_0^{\text{lit}}[0, \epsilon_2/2, \epsilon_3/2] &= \frac{\left[\left(\frac{Q}{2} + \epsilon_2 P_2 + \epsilon_3 P_3 \right)^2 - P_1^2 \right]^2}{4\epsilon_2 \epsilon_3 P_2 P_3 (Q + 2\epsilon_2 P_2) \times (Q + 2\epsilon_3 P_3)}. \end{aligned} \quad (54)$$

Here $T_0^{\text{H}} = L_0^{\text{lit}}$, as shown directly in Eq. (34). When the inserted label vanishes, the literature ratio is

$$\begin{aligned} L_0^{\text{lit}}[\epsilon_1/2, 0, \epsilon_3/2] &= \frac{1}{2} \left[P_2^2 - \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_3 P_3 \right)^2 \right], \\ \mathcal{B}_0^{\text{lit}}[\epsilon_1/2, 0, \epsilon_3/2] &= \frac{\left[P_2^2 - \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_3 P_3 \right)^2 \right]^2}{4\epsilon_1 \epsilon_3 P_1 P_3 (Q + 2\epsilon_1 P_1) \times (Q + 2\epsilon_3 P_3)}. \end{aligned} \quad (55)$$

Again $T_0^{\text{H}} = L_0^{\text{lit}}$. Finally, when the ket label vanishes, the literature ratio is

$$\begin{aligned} L_0^{\text{lit}}[\epsilon_1/2, \epsilon_2/2, 0] &= \frac{1}{2} \left[\left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 \right)^2 - P_3^2 \right], \\ \mathcal{B}_0^{\text{lit}}[\epsilon_1/2, \epsilon_2/2, 0] &= \frac{\left[\left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 \right)^2 - P_3^2 \right]^2}{4\epsilon_1 \epsilon_2 P_1 P_2 (Q + 2\epsilon_1 P_1) \times (Q + 2\epsilon_2 P_2)}. \end{aligned} \quad (56)$$

In this last case the human numerator has the opposite sign:

$$T_0^{\text{H}}[\epsilon_1/2, \epsilon_2/2, 0] = -L_0^{\text{lit}}[\epsilon_1/2, \epsilon_2/2, 0]. \quad (57)$$

The sign is fixed by the human BPZ and fixed-parity conventions; it is not a freely chosen phase. It disappears after squaring this numerator.

Three nonzero labels. All eight sign choices are covered by

$$\begin{aligned} L_1^{\text{lit}}[\epsilon_1/2, \epsilon_2/2, \epsilon_3/2] &= \frac{1}{2} \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b \right) \\ &\quad \times \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b^{-1} \right), \end{aligned} \quad (58)$$

$$\begin{aligned} \mathcal{B}_1^{\text{lit}}[\epsilon_1/2, \epsilon_2/2, \epsilon_3/2] &= - \frac{\epsilon_1 \epsilon_2 \epsilon_3 \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b \right)^2}{4P_1 P_2 P_3 (Q + 2\epsilon_1 P_1)(Q + 2\epsilon_2 P_2) \times (Q + 2\epsilon_3 P_3)} \\ &\quad \times \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b^{-1} \right)^2. \end{aligned} \quad (59)$$

The displayed L_1^{lit} is not yet identified with T_1^{H} . This is the first case in the 27-entry table where the convention translation must be checked using all terms in the tensor-product highest vectors. Equations (52)–(59) exhaust all 27 literature expressions $n_i \in \{0, \pm \frac{1}{2}\}$. They are the formulas evaluated by `Code/two_virasoro_fusion.py`; its unsquared output should therefore be read as L_a^{lit} , not yet as T_a^{H} .

8 Constructive physical- L_1 action: NS sector

This section determines the action of the *physical* NS Virasoro mode L_1 on every $\text{Vir}^{(1)} \oplus \text{Vir}^{(2)}$ branching primary. The auxiliary Majorana fermion is denoted by ψ , the physical free-field fermion by η , and all fermionic modes are in $\mathbb{Z} + \frac{1}{2}$.

8.1 The embedded generators, with the ψG term displayed

Set

$$\delta_b = b^{-1} - b, \quad U_m = \sum_{r \in \mathbb{Z} + \frac{1}{2}} \psi_{m-r} G_r. \quad (60)$$

For $m \neq 0$, Eq. (8) can be written as

$$\begin{aligned} L_m^{(1)} &= \frac{b^{-1}}{\delta_b} L_m - \frac{b^{-1} + 2b}{\delta_b} L_m^{\text{F}} + \frac{1}{\delta_b} U_m, \\ L_m^{(2)} &= -\frac{b}{\delta_b} L_m + \frac{b + 2b^{-1}}{\delta_b} L_m^{\text{F}} - \frac{1}{\delta_b} U_m. \end{aligned} \quad (61)$$

Thus neither embedded generator separately omits the ψG term. Two linear combinations that will be used below are the full operator identities

$$L_m^{(1)} + L_m^{(2)} = L_m + L_m^{\text{F}}, \quad (62)$$

$$bL_m^{(1)} + b^{-1}L_m^{(2)} = 2QL_m^{\text{F}} - U_m. \quad (63)$$

In Eq. (62) the two U_m coefficients cancel; in Eq. (63) the U_m term remains. This is the only cancellation used in the proof.

Let v_n , $n \in \frac{1}{2}\mathbb{Z}$, be an NS branching primary. Since both positive embedded modes annihilate it, Eq. (62) gives

$$(L_m + L_m^{\text{F}})v_n = 0, \quad L_m v_n = -L_m^{\text{F}} v_n, \quad m > 0. \quad (64)$$

The ψG term has not been discarded: it has canceled between the two embedded generators before Eq. (64) is applied.

8.2 Complex-fermion charge and the pair-field identity

Choose the reflected realization appropriate to the sign of n , and set

$$\chi_r = \psi_r - i\eta_r, \quad \bar{\chi}_r = \psi_r + i\eta_r, \quad \Psi_r = \frac{\chi_r}{\sqrt{2}}, \quad \Psi_r^* = \frac{\bar{\chi}_r}{\sqrt{2}}. \quad (65)$$

For $n \neq 0$, put $\sigma = \text{sgn}(n)$. The current is oriented branchwise so that the fermion appearing in the finite sea has charge σ :

$$[J_m, J_k] = m\delta_{m+k,0}, \quad [J_0, \Psi_r] = \sigma\Psi_r, \quad [J_0, \Psi_r^*] = -\sigma\Psi_r^*, \quad J_0 v_n = 2n v_n. \quad (66)$$

For the positive branch this is $J(z) = :\Psi(z)\Psi^*(z):$. For the negative reflected branch its negative is used. This convention keeps the finite-sea fermion equal to Ψ , while assigning the total charge $q = 2n$.

The finite product in Eq. (15), equivalently Eq. (4.22) of [arXiv:1608.02568](#), is exactly a charged complex-fermion vacuum:

$$v_n = \mathcal{N}_n |q = 2n; P\rangle, \quad J_m |q; P\rangle = 0 \quad (m > 0). \quad (67)$$

Its enlarged-module onset level is

$$N_n = 2n^2 = \frac{q^2}{2}. \quad (68)$$

Let L_m^η denote the physical-fermion stress tensor. The vector v_n contains no physical-boson oscillator, so the bosonic terms in the positive physical mode annihilate it and

$$L_1 v_n = L_1^\eta v_n. \quad (69)$$

Combining this with Eq. (64) gives

$$L_1 v_n = -\frac{1}{2}(L_1^F - L_1^\eta) v_n. \quad (70)$$

Direct substitution of Eq. (65) gives

$$L_m^F - L_m^\eta = \frac{1}{4} \sum_{r \in \mathbb{Z} + \frac{1}{2}} r (: \chi_{m-r} \chi_r : + : \bar{\chi}_{m-r} \bar{\chi}_r :). \quad (71)$$

The finite sea is made from Ψ , so its conjugate removes two sea fermions. Define the opposite-charge spin-two field

$$\mathcal{P}_\sigma^{\text{opp}}(z) = :\Psi^*(z)\partial\Psi^*(z): = \sum_m \mathcal{P}_{\sigma,m}^{\text{opp}} z^{-m-2}. \quad (72)$$

It has current charge -2σ . Antisymmetry implies

$$\mathcal{P}_{\sigma,m}^{\text{opp}} = - \sum_r r : \Psi_{m-r}^* \Psi_r^* :. \quad (73)$$

Equations (70) and (71) therefore give

$$\boxed{L_1 v_n = \frac{1}{4} \mathcal{P}_{\sigma,1}^{\text{opp}} v_n, \quad \sigma = \text{sgn}(n).} \quad (74)$$

The $\chi\chi$ term has charge $+2\sigma$ and would send q to $q + 2\sigma$, increasing $|q|$ by two. Its lowest possible energy is larger than the available energy $q^2/2 - 1$, so it annihilates v_n . The $\bar{\chi}\bar{\chi}$ term has charge -2σ and sends q to $q - 2\sigma$. The excitation level above the new charged vacuum is

$$d_n = \frac{q^2}{2} - 1 - \frac{(|q| - 2)^2}{2} = 2|q| - 3 = 4|n| - 3. \quad (75)$$

For $n = 0, \pm \frac{1}{2}$, no nonnegative target level is available and

$$L_1 v_0 = L_1 v_{1/2} = L_1 v_{-1/2} = 0. \quad (76)$$

For $|n| \geq 1$, the charge and level argument selects the adjacent charged vacuum and the relative level. The triangular-filtration argument below upgrades this to the double-Virasoro support statement

$$\boxed{L_1 v_n \in \text{span} \left\{ L_{-A}^{(1)} L_{-B}^{(2)} v_{n-\text{sgn}(n)} : |A| + |B| = 4|n| - 3 \right\}.} \quad (77)$$

8.3 Explicit Heisenberg descendant and normalization

Normalize a charge- α Heisenberg vertex by

$$V_\alpha(z)|q\rangle = z^{\alpha q} \exp \left(\alpha \sum_{m \geq 1} \frac{J_{-m}}{m} z^m \right) |q + \alpha\rangle \quad (78)$$

and define

$$\sum_{d \geq 0} H_d^{(\alpha)}(J_-) z^d = \exp \left(\alpha \sum_{m \geq 1} \frac{J_{-m}}{m} z^m \right), \quad dH_d^{(\alpha)} = \alpha \sum_{m=1}^d J_{-m} H_{d-m}^{(\alpha)}. \quad (79)$$

The mode $\mathcal{P}_{\sigma,1}^{\text{opp}}$ has vertex charge -2σ and selects the coefficient of z^{-3} , so Eq. (74) becomes

$$\boxed{L_1 v_n = \kappa_n^{\text{NS}} H_{4|n|-3}^{(-2 \text{sgn}(n))}(J_-) v_{n-\text{sgn}(n)}, \quad |n| \geq 1.} \quad (80)$$

The scale can be fixed without a Ward-identity calculation. For $k = 2n > 0$, choose the canonical charged vacuum

$$|q = k\rangle = \Psi_{-k+1/2} \Psi_{-k+3/2} \cdots \Psi_{-1/2} |0\rangle. \quad (81)$$

Reversing this order to the convention in Eq. (15), and using $\chi = \sqrt{2}\Psi$, gives

$$\mathcal{N}_{k/2} = 2^{k/2} (-1)^{k(k-1)/2} \Omega_k(P), \quad \frac{\mathcal{N}_{k/2}}{\mathcal{N}_{(k-2)/2}} = -2 \frac{\Omega_k(P)}{\Omega_{k-2}(P)}. \quad (82)$$

Choose the pair-field cocycle so that $\mathcal{P}_\sigma^{\text{opp}} = V_{-2\sigma}$. Then

$$\boxed{\kappa_n^{\text{NS}} = -\frac{1}{2} \frac{\Omega_{2n}(P)}{\Omega_{2n-2 \text{sgn}(n)}(P)}.} \quad (83)$$

The sign in Eq. (83) is checked independently by the first nonzero example below.

8.4 Triangular J_0 filtration and support theorem

The current zero mode does not commute with the two embedded Virasoro algebras. What is useful is a weaker, one-sided statement. Fix one transition $n \rightarrow n - \sigma$, $\sigma = \text{sgn}(n)$, and keep the reflected realization selected by the source label. Put

$$\hat{J}_0 = \sigma J_0, \quad [\hat{J}_0, \chi_r] = \chi_r. \quad (84)$$

Thus $\widehat{J}_0$ measures charge in the outward direction of the source Fermi sea.

For $m \neq 0$, define

$$S_m = L_m^{(1)} + L_m^{(2)}, \quad K_m = bL_m^{(1)} + b^{-1}L_m^{(2)}. \quad (85)$$

The first combination is the total stress tensor $S_m = L_m + L_m^F$, so it preserves the complex-fermion charge. Direct substitution of the free-field supercurrent into $K_m = 2QL_m^F - U_m$ gives

$$K_m = K_m^{(0)} + \frac{Q}{2}X_m^{(+2)}, \quad X_m^{(+2)} = \sum_{r \in \mathbb{Z} + \frac{1}{2}} r : \chi_{m-r} \chi_r :, \quad (86)$$

where

$$[\widehat{J}_0, S_m] = [\widehat{J}_0, K_m^{(0)}] = 0, \quad [\widehat{J}_0, X_m^{(+2)}] = 2X_m^{(+2)}. \quad (87)$$

There is no charge-lowering term. To see this directly, the c_j -dependent and momentum-dependent parts of U_m are modes of the neutral bilinear $:\psi\eta:$. The remaining part is

$$\begin{aligned} 2QL_m^F - iQ \sum_r r : \psi_{m-r} \eta_r : &= Q \sum_r r : \psi_{m-r} (\psi_r - i\eta_r) : \\ &= \frac{Q}{2} \sum_r r : (\bar{\chi}_{m-r} \chi_r + \chi_{m-r} \chi_r) :. \end{aligned} \quad (88)$$

The first term on the last line is neutral and the second is $X_m^{(+2)}$. Inverting the two linear combinations gives, for $\delta_b = b^{-1} - b$,

$$L_m^{(1)} = \ell_m^{(1)} - \frac{Q}{2\delta_b} X_m^{(+2)}, \quad L_m^{(2)} = \ell_m^{(2)} + \frac{Q}{2\delta_b} X_m^{(+2)}, \quad (89)$$

with both $\ell_m^{(i)}$ charge preserving. Hence every negative double-Virasoro mode preserves $\widehat{J}_0$ -charge or raises it by two. Equivalently,

$$[\widehat{J}_0, L_m^{(1)}] = -\frac{Q}{\delta_b} X_m^{(+2)}, \quad [\widehat{J}_0, L_m^{(2)}] = +\frac{Q}{\delta_b} X_m^{(+2)}. \quad (90)$$

In particular, $\widehat{J}_0$ neither commutes with the individual Virasoro generators nor exchanges the two algebras.

Let

$$v_- = v_{n-\sigma}, \quad q_- = 2|n| - 2, \quad d = 4|n| - 3 = 2q_- + 1. \quad (91)$$

Denote by $\mathcal{D}_{q_-}[d]$ the level- d double-Virasoro descendant space generated from v_- , and by $\mathcal{F}_{q_-}[d]$ the charge- q_- , level- d free-field space:

$$\begin{aligned} \mathcal{D}_{q_-}[d] &= \text{span} \left\{ L_{-A}^{(1)} L_{-B}^{(2)} v_- : |A| + |B| = d \right\}, \\ \mathcal{F}_{q_-}[d] &= \text{span} \left\{ c_{-\lambda} J_{-\mu} v_- : |\lambda| + |\mu| = d \right\}. \end{aligned} \quad (92)$$

By Eq. (89), a component of a vector in $\mathcal{D}_{q_-}[d]$ can only have charge $q_- + 2p$, $p \geq 0$. The operator $L_0^{(1)} + L_0^{(2)}$ is the ambient free-field grading, so every such component has relative energy d . The complex-fermion vacuum energy is $E_F(q) = q^2/2$. A component with $p \geq 1$ therefore requires at least

$$E_F(q_- + 2p) - E_F(q_-) = 2pq_- + 2p^2 \geq 2q_- + 2 = d + 1. \quad (93)$$

Only level d is available, so every charge-raised component vanishes. Consequently

$$\mathcal{D}_{q-}[d] \subseteq \mathcal{F}_{q-}[d]. \quad (94)$$

For generic b, P , the Fermi-sea construction and character proof of [arXiv:1111.2803](#), Sec. 3.2.1, give an injective tensor product of two Virasoro Verma modules generated by v_- . If $p(a)$ denotes the partition number, then

$$\dim \mathcal{D}_{q-}[d] = \sum_{a+b=d} p(a)p(b). \quad (95)$$

Boson–fermion correspondence gives the explicit basis in the second line of Eq. (92), and hence

$$\dim \mathcal{F}_{q-}[d] = \sum_{a+b=d} p(a)p(b). \quad (96)$$

These are concrete finite-dimensional subspaces of the same ambient Fock space. Thus Eq. (94) and equality of dimensions imply

$$\boxed{\mathcal{D}_{q-}[d] = \mathcal{F}_{q-}[d]}. \quad (97)$$

In particular the current descendant in Eq. (80) belongs to $\mathcal{D}_{q-}[d]$, proving Eq. (77) for every $|n| \geq 1$ at generic parameters. This proves existence of a double-Virasoro PBW expansion; it does not by itself give its coefficients. At $|n| = 1$ the target is the common uncharged vacuum v_0 , so no reflection convention is changed in this endpoint case.

8.5 Relative current–PBW reconstruction in the human-note conventions

We now turn the support statement into a finite algorithm for the two-Virasoro PBW coefficients. It is important not to obtain these coefficients by first writing a Shapovalov projection $\langle L_{-A}^{(1)} L_{-B}^{(2)} w_{n-1}, L_1 w_n \rangle$. Although that formula is formally correct, evaluating its left-hand side in this way requires either the unknown SCA-PBW expansion of w_n or an explicit reflected free-field bra. It therefore does not give an independent constructive algorithm.

This subsection uses exactly the notation and normalization of the human note. Thus

$$\chi_r = \psi_r - i\eta_r, \quad w_n(P) = \chi_{-(4n-1)/2} \chi_{-(4n-3)/2} \cdots \chi_{-1/2} |P\rangle, \quad (98)$$

with no factor of $1/\sqrt{2}$ in χ , while $\Psi = \chi/\sqrt{2}$ is used in the definition of the current J . An undecorated L_m is the physical SCA Virasoro mode. For $n \geq 0$, the normalized vector used elsewhere in this machine note and the raw human-note vector are related by

$$v_n = \mathfrak{s}_n(P) w_n, \quad \mathfrak{s}_n(P) = 2^{-2n} \ell(Q + 2P, 4n), \quad \mathfrak{s}_0(P) = 1. \quad (99)$$

We first state the construction for $n \geq 1$. Put

$$w_- = w_{n-1}, \quad q = 2n - 2, \quad d = 4n - 3 = 2q + 1. \quad (100)$$

Equation (80), translated with Eq. (99), becomes

$$\boxed{X_n := L_1 w_n = \kappa_n^{(w)} H_d^{(-2)}(J_-) w_-, \quad \kappa_n^{(w)} = \kappa_n^{\text{NS}} \frac{\mathfrak{s}_{n-1}(P)}{\mathfrak{s}_n(P)}}. \quad (101)$$

The scalar $\kappa_n^{(w)}$ depends on the raw-state ordering and the pair-field cocycle; the displayed conversion fixes it from the conventions already adopted above. The PBW direction below is independent of this overall scalar.

If $\mu = (1^{m_1} 2^{m_2} \dots)$ is a partition, set

$$J_{-\mu} = \prod_{r \geq 1} J_{-r}^{m_r}, \quad z_\mu = \prod_{r \geq 1} r^{m_r} m_r!, \quad \ell(\mu) = \sum_{r \geq 1} m_r. \quad (102)$$

The Heisenberg polynomial in Eq. (101) is then

$$H_d^{(-2)}(J_-) = \sum_{\mu \vdash d} \frac{(-2)^{\ell(\mu)}}{z_\mu} J_{-\mu}. \quad (103)$$

The two relative level- d bases are

$$D_{A,B} = L_{-A}^{(1)} L_{-B}^{(2)} w_-, \quad |A| + |B| = d, \quad (104)$$

$$F_{\lambda,\mu} = c_{-\lambda} J_{-\mu} w_-, \quad |\lambda| + |\mu| = d. \quad (105)$$

Both have cardinality

$$p_2(d) = \sum_{a=0}^d p(a) p(d-a). \quad (106)$$

At generic b, P , Eq. (97) says that they are bases of the same concrete subspace.

For positive-mode partitions define

$$c_\lambda = \prod_{r \geq 1} c_r^{m_r(\lambda)}, \quad J_\mu = \prod_{r \geq 1} J_r^{m_r(\mu)}. \quad (107)$$

Let $\text{Coeff}_{w_-}[Y]$ denote the coefficient of w_- after a relative level-zero state Y has been fully reduced. Define the terminal-reduction matrix

$$\boxed{K_{(\lambda,\mu),(A,B)} := \text{Coeff}_{w_-} [c_\lambda J_\mu L_{-A}^{(1)} L_{-B}^{(2)} w_-], \quad |\lambda| + |\mu| = |A| + |B| = d.} \quad (108)$$

This is an operator reduction, not a BPZ matrix element. In particular, it introduces neither a reflected bra nor the norm of w_- .

The defining current properties of X_n give the complete right-hand side. Since $c_m w_- = J_m w_- = 0$ for $m > 0$, the Heisenberg relations and Eq. (103) imply

$$\boxed{c_\lambda J_\mu X_n = \begin{cases} \kappa_n^{(w)} (-2)^{\ell(\mu)} w_-, & \lambda = \emptyset, \quad |\mu| = d, \\ 0, & \lambda \neq \emptyset, \quad |\lambda| + |\mu| = d. \end{cases}} \quad (109)$$

Consequently the desired human-note expansion

$$L_1 w_n = \sum_{|A|+|B|=d} C_{A,B}^{(n)} L_{-A}^{(1)} L_{-B}^{(2)} w_{n-1} \quad (110)$$

is determined by the square system

$$\boxed{\sum_{|A|+|B|=d} K_{(\lambda,\mu),(A,B)} C_{A,B}^{(n)} = \delta_{\lambda,\emptyset} \kappa_n^{(w)} (-2)^{\ell(\mu)}, \quad |\lambda| + |\mu| = d.} \quad (111)$$

Pure J -reductions alone give only $p(d)$ equations and are not sufficient once $d > 1$. The vanishing equations with $\lambda \neq \emptyset$ supply the remaining conditions.

The entries of K are computed recursively in the free-field realization of the embedded generators. For example,

$$\begin{aligned} J_m D_{A,B} &= [J_m, L_{-A}^{(1)} L_{-B}^{(2)}] w_-, \\ c_m D_{A,B} &= [c_m, L_{-A}^{(1)} L_{-B}^{(2)}] w_-, \end{aligned} \quad (112)$$

because the final positive mode annihilates w_- . Apply the Leibniz rule to the mode word, substitute Eq. (61), and normal order after every step relative to the charged vacuum w_- . No expansion of either w_- or w_n over the original SCA vacuum is needed.

The charge gap makes this reduction close on the basis (105). Indeed, Eq. (89) writes each embedded mode as a charge-preserving term plus a term that raises the outward current charge by two. Starting at charge q , the latter would require relative energy

$$E_F(q+2) - E_F(q) = 2q + 2 = d + 1, \quad (113)$$

whereas every intermediate state in the level- d reduction has at most energy d . All charge-raised terms therefore vanish identically, not merely after the final projection.

For completeness, one may instead construct the relative change-of-basis matrix T defined by

$$D_{A,B} = \sum_{|\lambda|+|\mu|=d} T_{(\lambda,\mu),(A,B)} F_{\lambda,\mu}. \quad (114)$$

Ordinary Heisenberg reduction gives

$$K_{(\lambda,\mu),(A,B)} = z_\lambda z_\mu T_{(\lambda,\mu),(A,B)}. \quad (115)$$

Thus invertibility of K follows from Eq. (97), and Eq. (111) is equivalent to coefficient matching against the known vector

$$x_{\lambda,\mu} = \delta_{\lambda,\emptyset} \kappa_n^{(w)} \frac{(-2)^{\ell(\mu)}}{z_\mu}. \quad (116)$$

At a Kac-degenerate value one performs the construction at generic b, P and then takes the limit, or works directly in the irreducible quotient.

8.6 First exact NS checks

The vanishing in Eq. (76) follows immediately from the explicit vectors $v_{\pm 1/2}$: $[L_1, G_{-1/2}] = G_{1/2}$, and the resulting positive mode annihilates the NS highest state.

For the first nonzero image, put $\sigma = \pm 1$ and

$$D_\sigma = 4P^2 + 8\sigma PQ + 3Q^2 + 4. \quad (117)$$

Applying physical L_1 to the explicit level-two vector recorded earlier gives, after cancellation of all $L_{-1}|P\rangle$ terms,

$$L_1 v_\sigma = -(Q + \sigma P) D_\sigma \psi_{-1/2} G_{-1/2} v_0. \quad (118)$$

Equation (63) is still a full operator identity. Only after acting on the vacuum branch may it be specialized:

$$L_{-1}^F v_0 = 0, \quad U_{-1} v_0 = \psi_{-1/2} G_{-1/2} v_0, \quad (119)$$

and hence

$$\left(bL_{-1}^{(1)} + b^{-1}L_{-1}^{(2)}\right) v_0 = -\psi_{-1/2} G_{-1/2} v_0. \quad (120)$$

Combining Eqs. (118) and (120) gives

$$\boxed{L_1 v_\sigma = (Q + \sigma P) D_\sigma \left(bL_{-1}^{(1)} + b^{-1}L_{-1}^{(2)}\right) v_0.} \quad (121)$$

This agrees with Eq. (80), since

$$\Omega_{2\sigma}(P) = \frac{1}{2}(Q + \sigma P)(Q + 2\sigma P)D_\sigma, \quad H_1^{(-2\sigma)} v_0 = -\frac{4}{Q + 2\sigma P} \left(bL_{-1}^{(1)} + b^{-1}L_{-1}^{(2)}\right) v_0. \quad (122)$$

The next case is already a level-three descendant of $v_{1/2}$:

$$L_1 v_{3/2} = \kappa_{3/2}^{\text{NS}} \left(-\frac{4}{3}J_{-1}^3 + 2J_{-1}J_{-2} - \frac{2}{3}J_{-3}\right) v_{1/2}. \quad (123)$$

Its two-*Virasoro* PBW expansion has ten candidate basis vectors, all inside the $v_{1/2}$ module. Producing their closed coefficients requires the same explicit Feigin–Fuchs oscillator rotation identified below in the Ramond analysis; it is not assumed in the support proof.

9 Ramond extension: status of the literature

This section records the literature status as of 20 August 2026. It is important to distinguish three increasingly strong statements:

1. the decomposition of an $F_0 \oplus \text{SCA}_0$ Ramond module into two-*Virasoro* modules;
2. a norm or Whittaker overlap of one two-*Virasoro* highest vector; and
3. the generic, chiral (NS, R, R) three-point coefficient, including its Ramond-ground-state phases.

Only the third statement supplies the coefficient required by the human note. In its notation the target is

$$\mathbb{B}_f^{(\eta)}(h_1, \beta_2, \beta_3; n_1, n_2, n_3) = \frac{\hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3})}{\|v_{1,n_1}\| \|v_{2,n_2}^{\alpha_2}\| \|v_{3,n_3}^{\alpha_3}\|}, \quad (124)$$

where $n_1 \in \frac{1}{2}\mathbb{Z}$, $n_2, n_3 \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}$, and the parity labels and the chiral sign η must be retained.

9.1 What has actually been established

Gauge-theory and block evidence. Ito’s ALE-space proposal, arXiv:1110.2176, identifies Ramond Whittaker norms and overlaps with instanton partition functions. Belavin–Mukhametzhanov, arXiv:1210.7454, similarly obtain four-point blocks containing Ramond fields from AGT data. These works provide strong independent control of Ramond blocks, but neither computes Eq. (124): their algebra and basis are not the present $\text{Vir} \oplus \text{Vir} \subset F \oplus \text{SCA}$ branching basis, and they do not determine its chiral three-point normalization.

The all-label double-Liouville candidate. Schomerus–Suchanek, arXiv:1210.1856, construct an infinite tower $\Phi_\alpha^{(k)}$ in super-Liouville theory times a free fermion. Integer k belongs to the NS sector and half-integer k to the Ramond sector. They propose that every $\Phi_\alpha^{(k)}$ is proportional to a primary of the two bosonic Liouville theories; this is precisely the local-field version of the branching-primary identification needed here. They give a candidate normalization and test several nontrivial local three-point functions, including Ramond labels $k = \pm\frac{1}{2}$ and $\pm\frac{3}{2}$.

This is the closest result in the literature to a formula for the full Ramond branching coefficient, but it is not yet Eq. (124). The paper explicitly does not prove the three-point identity for arbitrary labels. Its checks concern local, nonchiral correlators and leave a fourth root of unity $\varpi(k_3, k_2, k_1)$, with $\varpi^4 = 1$. Consequently it does not fix the separate holomorphic coefficient, the two Ramond parity copies, or the phases associated with a convention such as

$$G_0 w_\beta^\pm = i\beta e^{\mp i\pi/4} w_\beta^\mp. \quad (125)$$

It should be used below as an all-label candidate and a magnitude check, not as an already proved chiral branching formula.

The screening-charge proof stops in the NS sector. Hadasz–Jaskólski, arXiv:1312.4520, formulate the correspondence in terms of chiral structure constants and give a complete proof in the NS sector. Their reflection map, screening-charge matrix elements, singular-vector residues, and factorized chiral constants are the direct template for the calculation proposed below. They cite the Ramond tests of Schomerus–Suchanek and remark that a Ramond extension appears straightforward, but they do not carry out that extension. In particular, they do not introduce the Ramond zero-mode Clifford module needed to resolve the two parity copies in Eq. (124).

The Ramond module and highest vectors are known. Bershtein–Shchepochkin, arXiv:1608.02568, prove the generic Ramond branching rule

$$\mathcal{M}_R(\beta) \cong \bigoplus_{\epsilon=0,1} \bigoplus_{n \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}} \mathcal{V}_{c(1)}(h_n^{(1)}) \otimes \mathcal{V}_{c(2)}(h_n^{(2)}), \quad (126)$$

and give explicit highest vectors as finite products of integer modes $\chi_r^\pm = \psi_r - i\eta_r^\pm$, including the zero-mode factors that distinguish the two parity copies. Thus the vectorial input for a direct free-field calculation is available.

The same paper does *not* obtain generic three-point matrix elements. For the much weaker irregular/Whittaker overlap it states a conjectural squared coefficient in terms of the odd product s_{odd} . The conjecture was checked by computer for $|n| \leq 9/4$ (and in the reported series test through relative order ten), while its sign remains undetermined. This supplies a valuable norm/irregular limit check, but not the coefficient in Eq. (124).

10 Constructive physical- L_1 action: Ramond sector

This section gives the corresponding construction for the Ramond branching primaries. Here all fermionic modes are integers, $\delta_b = b^{-1} - b$, and $U_m = \sum_{r \in \mathbb{Z}} \psi_{m-r} G_r$. An undecorated L_m denotes the *physical* SCA mode, not either embedded Virasoro generator.

For $m \neq 0$, the complete embedded generators are

$$\begin{aligned} L_m^{(1)} &= \frac{b^{-1}}{\delta_b} L_m - \frac{b^{-1} + 2b}{\delta_b} L_m^F + \frac{1}{\delta_b} U_m, \\ L_m^{(2)} &= -\frac{b}{\delta_b} L_m + \frac{b + 2b^{-1}}{\delta_b} L_m^F - \frac{1}{\delta_b} U_m. \end{aligned} \quad (127)$$

Consequently

$$L_m^{(1)} + L_m^{(2)} = L_m + L_m^F, \quad (128)$$

$$bL_m^{(1)} + b^{-1}L_m^{(2)} = 2QL_m^F - U_m. \quad (129)$$

The ψG term is present in each generator. It cancels only in the first combination and remains explicitly in the second. No ψG term is omitted in what follows. The restriction to $m \neq 0$ avoids the Ramond normal-ordering constant, which is irrelevant for the $L_{\pm 1}$ argument.

If W_n^ϵ is a $\text{Vir}^{(1)} \oplus \text{Vir}^{(2)}$ highest vector, then

$$L_m^{(1)} W_n^\epsilon = L_m^{(2)} W_n^\epsilon = 0, \quad m > 0. \quad (130)$$

Equation (128), with its canceled U_m terms, then gives

$$L_m W_n^\epsilon = -L_m^F W_n^\epsilon. \quad (131)$$

10.1 Selection rule and exact low-level reductions

The onset level of the branch is

$$N_n = 2n^2 - \frac{1}{8}. \quad (132)$$

Thus the a priori expansion of $L_1 W_n^\epsilon$ may involve every $L_{-A}^{(1)} L_{-B}^{(2)} W_{n'}^\epsilon$ satisfying $N_{n'} + |A| + |B| = N_n - 1$. Direct free-field reduction instead gives the much sharper pattern

$$L_1 W_n^\epsilon \in \text{span} \left\{ L_{-A}^{(1)} L_{-B}^{(2)} W_{n-\text{sgn}(n)}^\epsilon : |A| + |B| = 4|n| - 3 \right\}. \quad (133)$$

The two parity copies do not mix. This pattern was first found by exact symbolic calculation through $|n| = 5/4$, and was checked at two generic rational points for $|n| = 7/4$. The free-field argument below first proves the all-label current-descendant formula and then proves the displayed double-Virasoro containment by a triangular J_0 -filtration.

At branch onset, $L_1 W_{\pm 1/4}^\epsilon = 0$. The first nonzero images are

$$L_1 W_{3\sigma/4}^0 = \frac{1}{4} W_{-\sigma/4}^0, \quad L_1 W_{3\sigma/4}^1 = \frac{1}{2} W_{-\sigma/4}^1, \quad \sigma = \pm 1. \quad (134)$$

In the PBW calculation it is essential to retain $[L_1, G_{-1}] = \frac{3}{2} G_0$. The result in Eq. (134) was also obtained independently by acting on the finite free-field χ -string.

The first genuine descendant example is the one at $n = 5/4$. Put $x = 2bP$. For either $\epsilon = 0, 1$,

$$\begin{aligned} L_1 W_{5/4}^\epsilon &= A L_{-2}^{(2)} W_{1/4}^\epsilon + B (L_{-1}^{(2)})^2 W_{1/4}^\epsilon + C L_{-1}^{(1)} L_{-1}^{(2)} W_{1/4}^\epsilon \\ &\quad + D L_{-2}^{(1)} W_{1/4}^\epsilon + E (L_{-1}^{(1)})^2 W_{1/4}^\epsilon, \end{aligned} \quad (135)$$

where

$$\begin{aligned}
A &= -\frac{x+b^2+6}{(x+b^2+4)(x+2b^2+1)}, & B &= \frac{2(2x+3b^2+7)}{(x+b^2+2)(x+b^2+4)(x+2b^2+1)}, \\
C &= \frac{8b^2}{(x+b^2+2)(x+2b^2+1)}, & D &= -\frac{b^2(x+6b^2+1)}{(x+b^2+2)(x+4b^2+1)}, \\
E &= \frac{2b^4(2x+7b^2+3)}{(x+b^2+2)(x+2b^2+1)(x+4b^2+1)}. & & (136)
\end{aligned}$$

The complete target space also contains level-two descendants of $W_{-1/4}^\epsilon$ and level-one descendants of $W_{\pm 3/4}^\epsilon$. Every one of those coefficients vanishes in the exact symbolic solve. Reflection gives the $n = -5/4$ result by $P \mapsto -P$ and $W_{1/4} \mapsto W_{-1/4}$.

For $n = 7/4$, Eq. (133) predicts twenty level-four descendants of $W_{3/4}^\epsilon$. All twenty occur. To record one fully exact benchmark, write

$$L_1 W_{7/4}^\epsilon = \sum_{|A|+|B|=4} \mathcal{C}_{A,B} L_{-A}^{(1)} L_{-B}^{(2)} W_{3/4}^\epsilon. \quad (137)$$

At $b = 3/2$, $P = 2/5$, the coefficient table is the same for both parity copies:

A	B	$\mathcal{C}_{A,B}$	A	B	$\mathcal{C}_{A,B}$
$\emptyset$	(4)	-2134586495/39213529488	(1, 1)	(2)	-17260875/355170424
$\emptyset$	(3, 1)	12777856925/365175993357	(1, 1)	(1, 1)	151880625/6615049147
$\emptyset$	(2, 2)	177475/10947384	(3)	(1)	13289200/135001003
$\emptyset$	(2, 1, 1)	-5338340750/365175993357	(2, 1)	(1)	-135049500/945007021
$\emptyset$	(1, 1, 1, 1)	379765000/365175993357	(1, 1, 1)	(1)	3999375/135001003
(1)	(3)	148525/2545011	(4)	$\emptyset$	-15271586235/78030579734
(1)	(2, 1)	-20210500/379206639	(3, 1)	$\emptyset$	1484874717075/8739424930208
(1)	(1, 1, 1)	3020000/379206639	(2, 2)	$\emptyset$	144803700/1345354823
(2)	(2)	4625800/57080961	(2, 1, 1)	$\emptyset$	-83962078875/624244637872
(2)	(1, 1)	-325624000/8505063189	(1, 1, 1, 1)	$\emptyset$	18105170625/1248489275744

The exact residual vanishes in the full auxiliary-fermion times physical Fock basis. The same closure was checked independently at $(b, P) = (5/3, 7/10)$. At total target level five there are also 102 basis vectors belonging to the other lower branch modules; none is needed to represent the image. This is the first check for which the sparsity is not a low-dimensional accident.

The reproducible implementation is the script `decompose_physical_l1.py` in the Ramond branching-check directory. For example, the benchmark above is generated by

```
python3 "python 2/ramond_branching_coefficient_check/decompose_physical_l1.py" \
--higher-label 7/4 --b 3/2 --momentum 2/5
```

10.2 Constructive free-field proof of the current-descendant formula

We now isolate which part of the preceding observation can be proved for all labels. All fermionic modes in this argument are Ramond integer modes. Choose the reflected free-field realization adapted to the sign of the branch label, suppress that sign from the notation, and introduce the second isotropic fermion

$$\chi_r = \psi_r - i\eta_r, \quad \bar{\chi}_r = \psi_r + i\eta_r, \quad \Psi_r = \frac{\chi_r}{\sqrt{2}}, \quad \Psi_r^* = \frac{\bar{\chi}_r}{\sqrt{2}}. \quad (138)$$

Then

$$\{\Psi_r, \Psi_s\} = \{\Psi_r^*, \Psi_s^*\} = 0, \quad \{\Psi_r, \Psi_s^*\} = \delta_{r+s,0}. \quad (139)$$

Put $\sigma = \text{sgn}(n)$. As in the NS construction, orient the current so that the finite-sea fermion has charge σ :

$$[J_m, J_k] = m\delta_{m+k,0}, \quad [J_0, \Psi_r] = \sigma\Psi_r, \quad [J_0, \Psi_r^*] = -\sigma\Psi_r^*. \quad (140)$$

For $n > 0$, this current is $J(z) =: \Psi(z)\Psi^*(z) :$; for $n < 0$, the negative of the reflected current is used.

There is an important convention at a sign-crossing transition. For each fixed transition $n \rightarrow n - \sigma$, the free-field realization and the orientation of J are chosen from the source label n and are then held fixed on both the source and target states. In particular, for $n = 3/4$ the target $W_{-1/4}^\epsilon$ is represented in the same current orientation as $W_{3/4}^\epsilon$. The reflected realization used to write a negative-label vector as a source of a different transition must not be introduced halfway through the argument: charges defined with two oppositely oriented currents cannot be compared.

The finite products in Eqs. (4.24)–(4.26) of [arXiv:1608.02568](#) say precisely that, in either parity copy, the branching vector is a charged complex-fermion vacuum:

$$W_n^\epsilon = \mathcal{N}_n^\epsilon |q = 2n; \epsilon, P\rangle, \quad J_m |q; \epsilon, P\rangle = 0 \quad (m > 0). \quad (141)$$

The two values of ϵ are two independent copies of the same charged Fock construction. The auxiliary and physical Ramond zero modes fix the normalization factors $\mathcal{N}_n^\epsilon$; they do not change the charge $q = 2n$.

Let L_m^η denote the physical-fermion part of L_m . Since W_n^ϵ contains no physical-boson oscillator, the positive physical mode satisfies

$$L_1 W_n^\epsilon = L_1^\eta W_n^\epsilon. \quad (142)$$

On the other hand, Eq. (131) gives $(L_1^F + L_1^\eta)W_n^\epsilon = 0$. Therefore

$$L_1 W_n^\epsilon = -\frac{1}{2}(L_1^F - L_1^\eta)W_n^\epsilon. \quad (143)$$

Direct substitution of Eq. (138) into the two Majorana stress tensors gives the operator identity

$$L_m^F - L_m^\eta = \frac{1}{4} \sum_{r \in \mathbb{Z}} r (:\chi_{m-r}\chi_r: + :\bar{\chi}_{m-r}\bar{\chi}_r:). \quad (144)$$

Thus the physical L_1 is reduced, on a branching highest vector, to a sum of two charge-changing fermion-pair modes.

This identity proves the triangular selection among charged free-fermion vacua. A charge q complex-fermion vacuum has diagonal fermion energy $q^2/2$. After applying a mode of index one, the energy is $q^2/2 - 1$. The $\chi\chi$ term has charge $+2\sigma$, so it would increase $|q|$ by two. Its lowest possible energy is too large, and it annihilates the state. The conjugate $\bar{\chi}\bar{\chi}$ pair has charge -2σ and sends q to $q - 2\sigma = 2(n - \text{sgn}(n))$. The excitation level over the new charged vacuum is

$$d = \frac{q^2}{2} - 1 - \frac{(|q| - 2)^2}{2} = 2|q| - 3 = 4|n| - 3. \quad (145)$$

For $|n| = 1/4$, this number is negative and the image vanishes. The pair operator is even, so it cannot mix the two parity copies.

The same argument gives an explicit current descendant rather than only its charge. Its conversion into the double-Virasoro descendant space is proved by the triangular filtration below. Define the opposite-charge spin-two field

$$\mathcal{P}_\sigma^{\text{opp}}(z) =: \Psi^*(z) \partial \Psi^*(z) := \sum_m \mathcal{P}_{\sigma,m}^{\text{opp}} z^{-m-2}. \quad (146)$$

It has current charge -2σ , and antisymmetry gives

$$\mathcal{P}_{\sigma,m}^{\text{opp}} = - \sum_r r : \Psi_{m-r}^* \Psi_r^* :. \quad (147)$$

Combining this with Eqs. (143) and (144) yields the exact branchwise operator formula

$$\boxed{L_1 W_n^\epsilon = \frac{1}{4} \mathcal{P}_{\sigma,1}^{\text{opp}} W_n^\epsilon, \quad \sigma = \text{sgn}(n).} \quad (148)$$

We next derive the conversion of the pair state into a current descendant directly from free-fermion commutators. This step is more than the particle-hole description and does not assume a bosonized vertex formula. Put

$$\alpha = -2\sigma, \quad q = 2n. \quad (149)$$

The current convention in Eq. (140) gives

$$[J_m, \Psi^*(z)] = -\sigma z^m \Psi^*(z). \quad (150)$$

When the commutator is differentiated, one apparently obtains an extra term proportional to $mz^{m-1} : \Psi^*(z) \Psi^*(z) :.$ This term vanishes by fermionic antisymmetry. The direct free-fermion commutator is therefore

$$\boxed{[J_m, \mathcal{P}_\sigma^{\text{opp}}(z)] = \alpha z^m \mathcal{P}_\sigma^{\text{opp}}(z).} \quad (151)$$

Let

$$F_q(z) := \mathcal{P}_\sigma^{\text{opp}}(z) |q; \epsilon, P\rangle. \quad (152)$$

For every $m > 0$, Eqs. (141) and (151) imply

$$J_m F_q(z) = \alpha z^m F_q(z). \quad (153)$$

On the other hand,

$$\left[J_m, \alpha \sum_{k \geq 1} \frac{J_{-k}}{k} z^k \right] = \alpha z^m. \quad (154)$$

It follows that

$$\exp \left(-\alpha \sum_{k \geq 1} \frac{J_{-k}}{k} z^k \right) F_q(z) \quad (155)$$

is killed by every positive current mode. It has charge $q + \alpha$, fixed physical momentum P , and the same Ramond copy ϵ . The Heisenberg Fock module has a unique highest vector with these data, hence

$$F_q(z) = \nu_\alpha z^s \exp \left(\alpha \sum_{m \geq 1} \frac{J_{-m}}{m} z^m \right) |q + \alpha; \epsilon, P\rangle \quad (156)$$

for a convention-dependent constant ν_α .

The power s is fixed without a vertex-operator assumption. The pair field has weight two, while a charge- q complex-fermion vacuum has weight $q^2/2$, up to the common Ramond normal-ordering constant. Therefore

$$s = \frac{(q + \alpha)^2}{2} - \frac{q^2}{2} - 2 = \alpha q, \quad \alpha^2 = 4. \quad (157)$$

Equations (156) and (157) are the usual charge-two boson-fermion formula, but here they have been obtained directly by solving the current commutator equations.

Introduce the universal polynomials

$$\sum_{d \geq 0} H_d^{(\alpha)}(J_-) z^d = \exp \left(\alpha \sum_{m \geq 1} \frac{J_{-m}}{m} z^m \right), \quad dH_d^{(\alpha)} = \alpha \sum_{m=1}^d J_{-m} H_{d-m}^{(\alpha)}. \quad (158)$$

Since $\mathcal{P}_\sigma^{\text{opp}}(z) = \sum_m \mathcal{P}_{\sigma,m}^{\text{opp}} z^{-m-2}$, the mode $\mathcal{P}_{\sigma,1}^{\text{opp}}$ is the coefficient of z^{-3} . The term $H_d^{(\alpha)}$ in Eq. (156) has power $z^{\alpha q + d}$. Consequently

$$\alpha q + d = -3, \quad d = -3 - \alpha q = 4|n| - 3. \quad (159)$$

This gives the exact current-descendant formula

$$\boxed{L_1 W_n^\epsilon = \kappa_n^\epsilon H_{4|n|-3}^{(-2 \text{sgn}(n))}(J_-) W_{n-\text{sgn}(n)}^\epsilon, \quad |n| \geq \frac{3}{4}.} \quad (160)$$

Using the constant $\nu_{-2\sigma}$ defined directly by Eq. (156), together with the normalization in Eq. (141), gives

$$\kappa_n^\epsilon = \frac{\nu_{-2 \text{sgn}(n)}}{4} \frac{\mathcal{N}_n^\epsilon}{\mathcal{N}_{n-\text{sgn}(n)}^\epsilon}. \quad (161)$$

Consequently the remaining overall factor is a finite-sea and zero-mode normalization, not an undetermined dynamical coefficient.

For the positive branch the first three polynomials needed here are

$$\begin{aligned} H_0^{(-2)} &= 1, & H_2^{(-2)} &= 2J_{-1}^2 - J_{-2}, \\ H_4^{(-2)} &= \frac{2}{3}J_{-1}^4 - 2J_{-1}^2 J_{-2} + \frac{1}{2}J_{-2}^2 + \frac{4}{3}J_{-1} J_{-3} - \frac{1}{2}J_{-4}. \end{aligned} \quad (162)$$

They correspond respectively to $n = 3/4, 5/4, 7/4$. In particular, $H_0 = 1$ explains why the first image is itself a lower branch primary, whereas H_2 and H_4 give the first genuine descendants.

Independent low-level check in the real-fermion basis. The script `check_ramond_l1_heisenberg.py` evaluates the two sides without using Eq. (156). On the positive branch it implements

$$J_m = i \sum_{r \in \mathbb{Z}} : \psi_{m-r} \eta_r : \quad (163)$$

directly in the auxiliary-fermion times physical-fermion Fock basis. It constructs $L_1 W_n^\epsilon$ by commuting the physical L_1 through the finite χ -string, constructs the polynomial $H_{4n-3}^{(-2)}(J_-) W_{n-1}^\epsilon$ independently, extracts the one overall proportionality coefficient, and checks every Fock coefficient in the residual. The exact results in the present finite-sea normalization are

n	ϵ	d	κ_n^ϵ	residual
3/4	0	0	1/4	zero (2/2)
3/4	1	0	1/2	zero (2/2)
5/4	0	2	1/2	zero (6/6)
5/4	1	2	1/2	zero (6/6)
7/4	0	4	1/2	zero (16/16)
7/4	1	4	1/2	zero (16/16)

Here (a/a) records the number of nonzero Fock coefficients on the two sides. The $n = 3/4$ test places both $W_{3/4}^\epsilon$ and the lower $W_{-1/4}^\epsilon$ in the same fixed current orientation; the current is not redefined when the branch label crosses zero. The check is run by

```
python3 "python 2/ramond_branching_coefficient_check/"\
"check_ramond_l1_heisenberg.py"
```

10.3 Triangular J_0 filtration and support theorem

The NS decomposition Eqs. (86)–(89) uses only the free-field formulas at nonzero mode number. It therefore carries over to Ramond integer modes. In the reflected realization selected by the source label, put $\hat{J}_0 = \sigma J_0$. Then, for every $m \neq 0$,

$$\begin{aligned} L_m^{(1)} &= \ell_m^{(1)} - \frac{Q}{2\delta_b} X_m^{(+2)}, & L_m^{(2)} &= \ell_m^{(2)} + \frac{Q}{2\delta_b} X_m^{(+2)}, \\ [\hat{J}_0, \ell_m^{(i)}] &= 0, & [\hat{J}_0, X_m^{(+2)}] &= 2X_m^{(+2)}, \end{aligned} \quad (164)$$

where now

$$X_m^{(+2)} = \sum_{r \in \mathbb{Z}} r : \chi_{m-r} \chi_r : . \quad (165)$$

The Ramond normal-ordering constant affects only mode zero and hence does not enter the descendant argument.

For $|n| \geq 5/4$, source and adjacent target use the same reflected realization. Set

$$W_-^\epsilon = W_{n-\sigma}^\epsilon, \quad q_- = 2|n| - 2, \quad d = 4|n| - 3 = 2q_- + 1. \quad (166)$$

Within a fixed parity copy define

$$\begin{aligned} \mathcal{D}_{q_-, \epsilon}[d] &= \text{span} \left\{ L_{-A}^{(1)} L_{-B}^{(2)} W_-^\epsilon : |A| + |B| = d \right\}, \\ \mathcal{F}_{q_-, \epsilon}[d] &= \text{span} \left\{ c_{-\lambda} J_{-\mu} W_-^\epsilon : |\lambda| + |\mu| = d \right\}. \end{aligned} \quad (167)$$

Equation (164) allows only charges $q_- + 2p$, $p \geq 0$, in the first space. The common Ramond vacuum constant cancels from energy differences, and

$$E_F(q_- + 2p) - E_F(q_-) = 2pq_- + 2p^2 \geq 2q_- + 2 = d + 1 \quad (p \geq 1). \quad (168)$$

Thus no charge-raised component fits at level d , and

$$\mathcal{D}_{q_-, \epsilon}[d] \subseteq \mathcal{F}_{q_-, \epsilon}[d]. \quad (169)$$

Theorem 4.2 of [arXiv:1608.02568](#) proves that, for generic momentum, each W_-^ϵ generates an irreducible and linearly independent tensor product of two Virasoro Verma modules. With the same partition-number notation as above, this gives

$$\dim \mathcal{D}_{q_-, \epsilon}[d] = \sum_{a+b=d} p(a)p(b) = \dim \mathcal{F}_{q_-, \epsilon}[d]. \quad (170)$$

The second equality follows from the two independent oscillator sets c_{-m} and J_{-m} ; fixing ϵ removes the Ramond ground-space multiplicity. The preceding inclusion is therefore an equality:

$$\boxed{\mathcal{D}_{q_-, \epsilon}[d] = \mathcal{F}_{q_-, \epsilon}[d]}. \quad (171)$$

Equation (160) then proves the all-label support rule (133) for $|n| \geq 5/4$.

The sign-crossing case $|n| = 3/4$ is a separate zero-mode base case: $d = 0$, and Eq. (134) identifies the image directly with the opposite $\mp 1/4$ primary in the same parity copy. Together with $L_1 W_{\pm 1/4}^\epsilon = 0$, this proves the Ramond support rule for every branch label at generic parameters. No assertion that J_0 is constant on a double-Virasoro module is used.

What remains constructive. The argument proves existence and uniqueness of the PBW expansion but does not give its coefficients. At level d , a direct algorithm is now well-defined:

1. expand the $p_2(d) = \sum_{a+b=d} p(a)p(b)$ ordered double-Virasoro PBW descendants in the fixed-charge basis $c_{-\lambda} J_{-\mu} W_-^\epsilon$;
2. invert this square generic matrix on the coefficient vector of $H_d^{(-2\sigma)}(J_-) W_-^\epsilon$;
3. multiply by the normalization κ_n^ϵ from Eq. (161).

Equation (171) proves that the matrix in the first step is invertible at generic parameters. A factorization into two Virasoro chain vectors could simplify this inversion, but is not needed for the support theorem. The remaining normalization problem is to evaluate the cocycle and Ramond zero-mode ratio in Eq. (161) for both parity copies.

11 Ramond literature verdict and remaining problem

The recent screening algebra does not close the gap. Ohkubo, [arXiv:2505.02926](#), constructs a q -deformation of $\mathbf{F} \oplus \text{SCA}$, two commuting q -Virasoro algebras, and screening currents whose degeneration gives the super-Virasoro screenings. The representation theory in that paper is NS. Its conclusion says that a modified Fock module and half-integer highest-weight charge should allow the Ramond sector; no Ramond branching coefficient is computed.

11.1 Verdict

No source found in the literature through the date above derives the generic chiral coefficient $\mathbb{B}_f^{(\eta)}$ for all branching labels while also fixing the two Ramond ground-state copies and their phases. The

present status is therefore

Ramond branching rule and explicit highest vectors: proved, all-label local double-Liouville formula: proposed and tested, generic chiral branching coefficient with phases: open.	(172)
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This conclusion is narrower than saying that Ramond blocks or Ramond super-Liouville three-point functions are unknown: both are known. What is missing is their convention-complete projection onto the specific two-*Virasoro* branching highest vectors.

11.2 Screening-charge route to be pursued here

The most economical derivation is now well localized:

1. Use the integer-mode products of arXiv:1608.02568 as the definitions of the two Ramond branching vectors $v_n^\epsilon(P)$, including their $\chi_0^\pm$ factors.
2. Extend the reflected-Fock-space BPZ pairing used in the NS calculation to the four-dimensional zero-mode ground space $F_0 \otimes SCA_0$. The pairing must implement Eq. (125), rather than identify the two parity copies implicitly.
3. Generalize the screening-charge matrix elements of arXiv:1312.4520 from NS exponentials to the spin-field intertwiners of an (NS, R, R) vertex. Charge neutrality fixes the number of screenings; the zero-mode Clifford matrix fixes which of the two chiral Ramond structures is selected.
4. Evaluate the resulting finite Selberg-type products and divide by the three reflected-Fock norms. The product must reduce to the Schomerus–Suchanek double-Liouville candidate after multiplying the two chiral halves and forgetting the zero-mode phase.
5. Anchor the remaining square-root choices at $n_2, n_3 = \pm \frac{1}{4}$ by a direct ground-state Ward-identity calculation, then propagate them by the screening shifts. In the Whittaker limit the square must reproduce the s_{odd} formula conjectured and tested in arXiv:1608.02568.

The first new calculation should therefore be the four-by-four matrix of ground-level (NS, R, R) branching numerators. It will separate a genuine dynamical product formula from the convention-only zero-mode phases before higher integer χ_{-r} modes are introduced.

12 What remains to be audited

The calculation above settles the all-NS coefficient, modulo a direct audit of the literature product formula against the human note’s ordered Ward identities. The efficient audit is:

1. reproduce Eqs. (41) from the Gram form;
2. reproduce Eqs. (46) and Eq. (47) from the three-point Ward identities;
3. check a $|k| = 2$ case using the explicit level-two vectors in Appendix B of arXiv:1111.2803; and

4. only then use Eq. (43) as the all-level input to the block decomposition.

The Ramond branching vectors have an analogous free-field prescription in [arXiv:1608.02568](#). It uses integer $\chi_r^\pm$ modes and has two parity copies. Their norms and (NS, R, R) three-point coefficients require a separate translation to the human note's phase convention

$$G_0 w_\beta^\pm = i\beta e^{\mp i\pi/4} w_\beta^\mp. \quad (173)$$

No Whittaker coefficient is used here.

13 Sources

The free-field vector, its PBW normalization, the blow-up factors, and the functions $s_{\text{even}}, s_{\text{odd}}$ are in [arXiv:1111.2803](#), especially Sec. 3.2, Proposition 3.2, and Appendix B. The later presentation and the Ramond analogue are in [arXiv:1608.02568](#), Sec. 4.

The Ramond local-field candidate and its checked three-point examples are in [arXiv:1210.1856](#), Secs. 4.2–4.3. The chiral screening/reflection proof that presently exists only in the NS sector is [arXiv:1312.4520](#). Indirect Ramond Whittaker and block evidence comes from [arXiv:1110.2176](#) and [arXiv:1210.7454](#). The recent q -deformed algebra and screening-current construction, which leaves its Ramond representation theory for future work, is [arXiv:2505.02926](#).